

14.15 Solution: (a) Since the motion is axially symmetric, the result will only depend on the azimuthal angle θ . Without loss of generality, we can choose $\mathbf{n}$ to be in the xz -plane, *i.e.*, $\mathbf{n} = (\sin \theta, 0, \cos \theta)$. Then, with $\mathbf{x}(t) = R(\cos(\omega_0 t), \sin(\omega_0 t), 0)$ and $\mathbf{v}(t) = \dot{\mathbf{x}}(t) = \omega_0 R(-\sin(\omega_0 t), \cos(\omega_0 t), 0)$, we have

$$\mathbf{n} \cdot \mathbf{x}(t) = R \sin \theta \cos(\omega_0 t), \quad \mathbf{v}(t) \times \mathbf{n} = \omega_0 R(\cos \theta \cos(\omega_0 t), \cos \theta \sin(\omega_0 t), -\sin \theta \cos(\omega_0 t)).$$

Applying the result of Prob. 14.13,

$$\frac{dP_m}{d\Omega} = \frac{e^2 \omega_0^4 m^2}{(2\pi c)^3} \left| \int_0^{2\pi/\omega_0} \begin{pmatrix} \cos \theta \cos(\omega_0 t) \\ \cos \theta \sin(\omega_0 t) \\ -\sin \theta \cos(\omega_0 t) \end{pmatrix}^\top \omega_0 R \exp \left(im\omega_0 \left[t - \frac{R}{c} \sin \theta \cos(\omega_0 t) \right] \right) dt \right|^2.$$

Since

$$\exp \left(-im\omega_0 \frac{R}{c} \sin \theta \cos(\omega_0 t) \right) = \sum_{k=-\infty}^{\infty} (-i)^k J_k \left(m \frac{\omega_0 R}{c} \sin \theta \right) e^{-ik\omega_0 t},$$

the integral can be evaluated with the following identities,

$$\begin{aligned} & \int_0^{2\pi/\omega_0} \begin{pmatrix} \cos(\omega_0 t) \\ \sin(\omega_0 t) \end{pmatrix}^\top \exp(im\omega_0 t) \sum_{k=-\infty}^{\infty} (-i)^k J_k(m\beta \sin \theta) e^{-ik\omega_0 t} \\ &= \int_0^{2\pi/\omega_0} \begin{pmatrix} \frac{1}{2}(e^{i\omega_0 t} + e^{-i\omega_0 t}) \\ \frac{1}{2i}(e^{i\omega_0 t} - e^{-i\omega_0 t}) \end{pmatrix}^\top \exp(im\omega_0 t) \sum_{k=-\infty}^{\infty} (-i)^k J_k(m\beta \sin \theta) e^{-ik\omega_0 t} \\ &= \frac{2\pi}{\omega_0} \begin{pmatrix} \frac{1}{2}((-i)^{m+1} J_{m+1}(m\beta \sin \theta) + (-i)^{m-1} J_{m-1}(m\beta \sin \theta)) \\ \frac{1}{2i}((-i)^{m+1} J_{m+1}(m\beta \sin \theta) - (-i)^{m-1} J_{m-1}(m\beta \sin \theta)) \end{pmatrix}^\top \\ &= \frac{2\pi}{\omega_0} \begin{pmatrix} \frac{1}{2}(-i)^{m+1} \cdot 2 \frac{dJ_m(x)}{dx} \Big|_{x=m\beta \sin \theta} \\ \frac{1}{2}(-i)^{m+2} \cdot \frac{2m}{m\beta \sin \theta} J_m(m\beta \sin \theta) \end{pmatrix}^\top. \end{aligned}$$

Then,

$$\begin{aligned} \frac{dP_m}{d\Omega} &= \frac{e^2 \omega_0^4 R^2}{2\pi c^3} m^2 \left| \begin{pmatrix} \cos \theta \frac{dJ_m(x)}{dx} \Big|_{x=m\beta \sin \theta} \\ (-i) \cos \theta \frac{J_m(m\beta \sin \theta)}{\beta \sin \theta} \\ -\sin \theta \frac{dJ_m(x)}{dx} \Big|_{x=m\beta \sin \theta} \end{pmatrix}^\top \right|^2 \\ &= \frac{e^2 \omega_0^4 R^2}{2\pi c^3} m^2 \left\{ \left[\frac{dJ_m(m\beta \sin \theta)}{d(m\beta \sin \theta)} \right]^2 + \frac{\cot^2 \theta}{\beta^2} J_m(m\beta \sin \theta)^2 \right\}. \end{aligned}$$

(b) In the non-relativistic limit, $\beta \ll 1$, only $m = 1$ component will contribute. Notice that

$$J_1(z) \sim \frac{z}{2}, \quad \frac{dJ_1(z)}{dz} \sim \frac{1}{2},$$

we will arrive at the following asymptotic result,

$$\frac{dP}{d\Omega} \sim \frac{dP_1}{d\Omega} = \frac{e^2 \omega_0^4 R^2}{2\pi c^3} \left(\frac{1}{4} + \frac{1}{4} \cos^2 \theta \right) = \frac{e^2 \omega_0^4 R^2}{4\pi c^3} \left(1 - \frac{1}{2} \sin^2 \theta \right),$$

which agrees with Prob. 14.4 (b).